



MECH MAINT- BOILER Department
SIMHADRI

Note No :1

Ref:SMTTPP/MECH MAINT-/2023-24/BMD/RF/436379

Date:

29/08/2023(12:14:03)

Subject:Revised LMI of Guidelines for Inspection and Maintenance for Radial Fans

The LMI on "Guidelines for Inspection and Maintenance for Radial Fans." LMI/OGN/OPS/MECH/013/ has been revised as per revised OGN "COS-ISO-00-OGN-OPS-MECH-13; Rev. No.: 3 Aug 2023.".

Revised LMI and OGN is attached.

Put up for kind approval please.

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Note No:6

LMI has been reviewed and found that Monitoring of Moisture, MI and WDA are being in practice as per the LMI.

May please forwarded for further approval.

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Note No:9

Pl review the LMI w.r.to the OGN

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Note No:11

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Note No:13

LMI has been reviewed and pls be ensure the DPT/MPI/UT shall be carried out with ASNT/ISNT level-II certified personel as per the requirement.

May pls forward for further approval.

Submitted by:

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Note No:15

LMI has been revised as per revision in revised OGN which is given in clause no.3.0 on page no. 4 of the revised LMI as given/attached in annexure for approval. Clause no. 5.5 (as per revised OGN) is NA as it for "Servicing of Variable Speed Drive -Voith/Fluid coupling (If applicable)".

Submitted for kind approval of C.A. (HOP Simhadri).

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NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

LMI/OGN/OPS/MECH/013/

ISSUE NO.1: REVISION NO. 7.00: DATE: AUG.'2023

Guidelines for Inspection and Maintenance for Radial Fans

Approved for
Implementation by -----
HOP (Simhadri)

Date: -----

Enquiries to: GM (Operation & Maintenance)

LMI APPROVAL DOCUMENT

1	TITLE OF LMI	Guidelines for Inspection and Maintenance for Radial Fans.
2	DOCUMENT NUMBER REVISION/ ISSUE NO & DATE	<u>LMI/OGN/OPS/MECH/013/ ISSUE-1; REV. - 7.00; AUG.'2023</u>
3	ORIGINAL/ REF DOCUMENT	COS-ISO-00-OGN-OPS-MECH-13 Rev. No.: 3 Aug 2023
4	PREPARED BY	SIGNATURE: Ritesh Kumar, Sr. Mgr(BMD)
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8	APPROVED FOR ISSUE	SIGNATURE: S.K Sinha, HOP(SSTPP)
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Guidelines for Inspection and Maintenance for Radial Fans

INTRODUCTION

The reliability, efficiency, and safety of draft fans in coal based thermal power plants depend on effective operating and maintenance practices. This document aims at providing adequate guidelines for safe operation and maintenance practices for Radial Fans. Radial fans are used in various application for handling flue gas, air & hot air in power plants. They may have either journal/ sleeve bearings or anti friction bearings. These fans require periodic inspection and maintenance to ensure their mechanical and aerodynamic integrity for reliable & efficient operation as well as safety precautions. Periodic testing of lub oil, lesser vibrations & lesser ash content in flue gas are essential for longer life & trouble-free operation of fan. The OEM maintenance recommendations should also follow for each fan in order to track reoccurring damage mechanisms as well as to prevent them. The maintenance areas include impeller & shaft inspection, shaft bearings, inlet guide vanes and fluid coupling.

NTPC Simhadri is having KKK (BHEL) make Radial Fans:

2.0 SUPERSEDED DOCUMENTS

Location Management Instruction - Inspection Requirements of Radial ID Fans of BHEL Make for 500MW Units - LMI/OPS/MECH/013 Rev. No.: 6 Dated: Aug' 2023

3.0 REASON FOR REVISION

Following changes done as per latest revision of OGN in Aug'23

Clause 5.1.5: Digital drawing point for spare part included.
Clause 5.1.13: Points regarding readiness of spare assembly added.
Clause 5.1.14: Mismatch in ID Fan currents included.
Clause 5.1.15: Removal of broken bolts in assemblies included.
Clause 5.2.4: Rope type added.
Clause 5.3.5: Journal bearing Thrust collar bolts tightness clause included.
Clause 5.4.4: Checking of Relief valves / PRV included.
Clause 5.7.6: Check of IGV link assembly added.
Clause 5.7.7: Components under Fatigue loading
Clause 8.0: T&P list updated.
Clause 9.0: Merged with clause 5.1.6

4.0 SAFETY ASPECT

- 4.1 Permit to work has to be obtained before start of the work by competent person. The permit must be signed by both SAP & CP and PTW with lock out key has to be kept at work site.
- 4.2 Pre talk of worker before starting the maintenance, discuss all the activities to be carried out/covered in job safety analysis (JSA) & discuss the risks associated with them.
- 4.3 All personnel that operate, inspect, or maintain the fans should be trained in the plant safety procedures.
- 4.4 The fan main motor and fan auxiliaries must be electrical isolated & locked.
- 4.5 The respective discharge damper must also be in closed condition & electrically isolated, if working while other fan is in running condition.
- 4.6 The fan parts should be lifted in the hoisting lugs intended for that purpose.
- 4.7 Proper illumination at workplace is important to ensure safe working.

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4.8 Even when a fan is locked out electrically, it may be subject to “wind milling” because of pressure differential across the fan. Therefore, the impeller should be secured to physically

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restrict rotational movement, if required.

4.9 Inspection doors must not be removed until the rotor is stationary. Before opening the manhole door, ensure fan duct/ housing is not in pressurized condition because of any reason.

4.10 Flue gas handling fans are confined spaces that may be filled with harmful gases. Confined space entry procedures should be followed before entry.

4.11 Depressurize hoses/pipes containing high-pressure fluids and allow them to cool down to a suitable temperature.

4.12 Use extreme caution when handling hot objects like couplings to avoid personal injury.

4.13 If an oil bath is used for heating any object, the oil must have a flash point of 350°F (177°C) or greater.

4.14 Mandatory use of safety belts/ harness with life line by workers working at fan casing/ height with proper scaffolding.

4.15 Immediately clean the lube oil spills properly during the work to avoid injury due to slippage.

4.16 All coupling guards must be fitted in position before cancellation of PTW.

4.17 Before cancellation of PTW, it should be ensured that all personnel, scrap and tools must be removed from the fan.

4.18 Vibration and performance testing require the fans to be running during the tests. Special precautions must be observed while performing these tests.

4.19 Loose cloths that may be sucked/ rolled on the fan rotating parts, must be avoided.

4.20 The healthiness of earthing system of motor must be ensured with help of electrical department for safety of man & machine.

4.21 The use of 24-volt supply hand lamps shall be ensured. If available, use DC lights for working inside the internal chamber of fan during breakdown/short shutdown.

4.22 Load testing of lifting tools & tackles, supporting structures by Safety / FQA / third party & readiness before usage during maintenance work.

4.23 Do not weld on any part of fan unless the welding machine is grounded to the respective part to prevent arcing across bearing assemblies.

5.0 MAINTENANCE PRACTICES

The following maintenance practices are to be adopted for radial fans. Overhauling protocols / critical dimension are to be recorded / maintained as per OEM recommendations. Obtain & maintain the records/protocols of original dimensions during erection/commissioning stage in newly commissioned fans for reference.

General Guidelines

5.1.1 Record the important parameters of fan like vibration level on all bearings of fan & motor, bearing temperature, motor amperage, lube oil tank temperature; Lube oil pumps discharge pressure, etc before stopping the Fan. Check the recent wear debris analysis report also.

5.1.2 As per BS ISO 10816-3, the fan bearing vibrations should be limited to less than 2.3 mm/sec for new fans being commissioned and after overhaul/repair or OEM recommended, whichever is less. Signature analysis is to be done to find out root cause of vibrations. If vibrations are ≥ 4.5 mm/s suitable corrective action to be taken.

Guidelines for Inspection and Maintenance for Radial Fans

5.1.3 After taking permit to work, the Fan motor Coupling alignment readings may be recorded/ ensured for reference during re-commissioning.

5.1.4 Ensure availability of spares, consumables, special tools, welding machine, gas cutting set, service air point & readiness of hoist.

5.1.5 Digital drawing to be developed for emergency requirement spares, which are not available in stock.

5.1.6 In new units, Equipment being critical / precision in nature, Technical assistant services of the OEM should be explored at least for the maintenance / OH of rotary equipment for reliability / availability of equipment and also to familiarize fan maintenance personnel with the new design, best practices, usage of T&P and spare parts planning. The technical advisor (TA)/ OEM/OES Expert visit report of fan trouble shooting consisting dismantling / assembling procedure & critical clearances etc, is to be prepared and shared on BMD dashboard, RP KT meet and COS Boiler. The history of fan maintenance must be maintained & should kept on BMD dashboard at plant webpage. The root cause analysis of each breakdown is to be carried out. The C-2 form for Rotary Machines must be filled & sent to CC-OS after any breakdown, which can repeat, caused generation loss or critical component failure. The health matrix (as given in annexure) of fans must be filled on quarterly basis & sent to CC-OS. 5.1.7 The following activities must be planned in every fan overhaul-

- Coupling inspection & alignment
- Impeller & shaft assembly Inspection
- Bearing Inspection (DPT & UT)
- Lube oil system servicing & tank cleaning
- Servicing of inlet guide vanes
- Servicing of Variable Speed Drive (Fluid coupling)
- Inspection of Silencer, Suction chamber and Spiral casing for any cracks, corrosion & looseness.
- Other activities as per requirement / OEM recommendation
- Shaft Critical welding Inspection through DPT/MPI/UT

5.1.8 Inspect any oil leakage from bearing housings / lube oil system of fan.

5.1.9 The spare assembly of radial fans should be stored indoor & critical portions should be properly covered with coating/ grease. The impeller assembly should be rotated by 90° in every three months.

5.1.10 Marking the assembled components by match marks before dismantling.

5.1.11 Cleaning of fan after maintenance for foreign materials like loose bolts, nuts, welding electrodes, any debris, etc.

5.1.12 During re-commissioning of the units after overhaul, coupled trial run of fans should be taken before boiler light-up in presence of Maintenance and Operation personal. Line up of lubrication and duct system to be ensured before trial run.

5.1.13 All the used spare assembly in fans must be made available as soon as possible

5.1.14 If ID Fans are taking less than rated current or any mismatching of current at full

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load, then fan internals, IGV setting, cone clearances etc to be checked & attended during opportunity.

5.1.15 In all fans, all parts/ sub-assemblies must be tightened with required number of bolts (as given in design). Any broken bolt must be removed.

5.1.16 The foundation spring isolator units do not require any special maintenance. Routine inspection of the surface should be carried out for corrosion /painting/ damages. Spring units should be cleaned by air (Not by water). A record of spring height wrt original settings during erection/commissioning stage/as per drawing may be recorded/verified during overhauling, if available. In case of any major load (Machine Parts) to be removed from foundation, the pre stressing bolts shall be hand tightened and later should be released after keeping the load on foundation. OEM of spring isolator to be called during capital overhaul for spring check.

5.2 Impeller & Shaft assembly Inspection

5.2.1 Impeller conical cover plates are required to be dismantled (if removable) for inspection / NDT (DPT/MPI/UT) of welds of shaft flange and shaft rings to the shaft.

5.2.2 Visual inspection of all shaft and impeller welds during major overhauls including the balance weight welds. They should be inspected for any signs of obvious distress and to assess the extent of any wear. There is no formal procedure for visual inspection. The results of inspection should be recorded.

5.2.3 If there is any vibration due to ash ingress into the impeller cone, then also ash to be removed only by dismantling the impeller cone. Pocket cutting in cone is not permitted for any purpose.

5.2.4 Every time, whenever cones are dismantled, rope fixed between shaft ring and cone ring should be replaced with adequate size. Suitable sealant may be used with rope to prevent ash ingress. Ceramic rope may be used, if having frequent ash ingress problem in impeller cone.

5.2.5 Cone bolts are to be torque tightening as per drawing. Ensure tightness of all critical fasteners.

5.2.6 All the shaft welds should be inspected for defects like cracks, undercuts, notches, etc. by NDT (Magnetic Particle Inspection/Liquid Penetrant Testing). Prior to the inspection, the location of any significant defects known from previous inspection should be brought to the attention of the NDT operator so that the progress of such defects can be most carefully monitored. The welds and heat affected zones should be properly cleaned for carrying out the NDT checks.

5.2.7 If notches and undercuts defects observed in visual inspection/ NDT, they should be ground smoothly prior to return to service. No surface cracks either in welds or heat affected zones are permitted for any cracks observed. Detailed investigation is required and the fan should not be taken into operation. Manufacturer and Corporate Operation should be informed for their action.

5.2.8 The thickness of critical impeller plates like center plate, side plates, blades and blade liners are to be measured during overhaul & compare it by original thickness.

5.2.9 If welding is required on impeller or shaft due to severe erosion or any other reasons,

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OEM recommendations must be followed after getting detailed repair procedure.

5.3 Inspection of bearings

Radial fans may be provided with either sleeve bearings (journal bearings) or anti friction bearings. The sleeve bearings may be with or without cooling water jacket and oil rings

5.3.1 Determine bearing condition by operating conditions, measuring bearing operating temperature, or by using a predictive maintenance technique, such as vibration analysis or oil analysis.

5.3.2 In case sleeve bearings, An inspection should be made of the babbitt surface for surface scoring or wiping (axial scoring is unacceptable; circumferential scoring is not to exceed the manufacturer's recommendations), Fatigue cracking, Corrosion, Pitting due to electrical discharge, Overheating, Uneven wear, Fretting etc.

5.3.3 Dye penetrant test or Ultrasonic testing of sleeve bearings (lower & upper half) must be carried out for checking cracks & bonding of Babbitt metal.

5.3.4 An inspection should be made of the shaft journal for ridges, grooves, and/or sharp edges, and, if they are present, they should be removed with abrasive cloth.

5.3.5 Thrust collars should be inspected for axial and radial run out. On a thrust bearing, the face-to-face dimension is critical. Measure the clearance between the sleeve bearing thrust face and the thrust collar. It should be within limit as per drawing. Journal bearing Thrust collar bolts tightness to be checked at specified torque.

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5.3.6 Clearance & blue match of Journal/ sleeve bearings with shaft journals must be checked as per procedure given by OEM.

5.3.7 Seals of bearing housing should be replaced in every overhaul.

5.3.8 Inspect bearing oil rings (if provided) for damage or looseness of joint.

5.3.9 Special care should be taken for inlet & outlet water connection of bearing cooling jacket (if provided) and should test after charging cooling water line before tightening upper housing part. Thread sealant compound may also be used.

5.3.10 Keep the bearings clean from dust/ ash & lubricate the bearings during box up & thoroughly cleaning of bearing housings is also mandatory.

5.3.11 In case of sleeve bearing replacement, blue match the spherical portion of bearing with the housing /bearing support.

5.3.12 It is suggested to replace bearings of seal air fans in every overhaul.

5.4 Servicing of Lube oil System

5.4.1 Thorough cleaning of lube oil tank should be carried out by good quality cloth. Never clean the tank with the help of cotton waste, it may cause frequent filter choking problem. Any rusting of tank to be removed and coated with rust preventive painting. All door gaskets of sump/ tank should be inspected and replaced as necessary.

5.4.2 Inspection, hydraulic testing & replacement of O-rings /gaskets of lube oil coolers must be done after cleaning of tube bundle.

5.4.3 The breather /cap of lube tank must be cleaned & in position. All the view glasses in lube oil system should also be cleaned. View glasses should be provided in individual bearing return oil lines, if not available.

5.4.4 All lube oil lines should be flushed & clean. Attend all lube oil leakages & other defects based on the requirement. Relief valves / PRV to be checked for any abnormality.

5.4.5 Filter/ strainer cleaning, inspection & replacement based on condition. If there are paper filters in the system then these must be replaced, never try to clean the paper filters. Also maintain sufficient stock of the paper filters.

5.4.6 Ensure correct grade of lube oil & its moisture content in case of new oil before use. A permanent board (lubrication chart) should be displayed in lube oil storage area at site, showing use of lube oil/ grease grade equipment/ subassembly wise.

5.4.7 Never fill the oil more than maximum level.

5.4.8 Monthly oil monitoring for moisture & mechanical impurities (MI) and Quarterly oil monitoring through wear debris analysis (WDA) must be followed strictly. The safe limit of PQ index for fans is 25 (As given in Netra WDA report). The moisture content should be less than 100 PPM or as recommended by OEM. Oil should be clear and transparent and should not be hazy. Centrifuging may be done to remove moisture. In case of any abnormality in the WDA, necessary corrective action as follows should be taken. Oil sample should be collected from system before filters.

Note:- If PQ index is beyond the safe limit. Bearing inspection must be planned in next opportunity. However, partial oil replacement to be done immediately with precautions as a temporary measure and sample may be taken after 7 days to resend for WDA to assess

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the speed of bearing deterioration. If this report is also beyond acceptable limits, bearing replacement to be planned at the earliest.

5.4.9 Use correct pressure rating & length of oil hoses for lube oil/ cooling water systems. Also prevent rubbing of hoses to each other or other surfaces. Ensure fixing of protective sheath of proper grade in case rubbing can't be avoided. The cooling water & oil hoses should be inspected for physical damage.

5.4.10 Check the vibrations in various lube oil lines & use additional supports with rubber pads specially fan equipped with fluid coupling.

5.4.11 During re-commissioning of lube oil system, one must ensure that oil is coming back to tank from all return oil lines. All system components and lines should be primed and vented.

5.5 N.A

5.6 Coupling inspection & alignment

5.6.1 Visually inspect the coupling for wear, corrosion, cracking after cleaning.

5.6.2 In case of lubricated couplings, Dis-assemble, clean, and replace lubricant & Verify that the proper lubricant is being used. Inspect gaskets and O-rings for elasticity & replace if required.

5.6.3 Inspect the flexible member like grid, spring, rubber bushes or coupling teeth in case of gear couplings.

5.6.4 Verify the magnetic center of motor during decoupled trial run to prevent the thrust loading on the motor thrust bearing due to fan. This activity is to be checked in case of motor having journal bearings.

5.6.5 Perform precision shaft alignment as per OEM recommendation after maintaining correct coupling gap. Must use calibrated alignment instruments like dial gauge.

5.6.6 While replacing coupling, proper interference fit must be ensured as per OEM recommendations. It is important when mounting interference coupling hubs to make sure that clearance exists over the top of keys: otherwise, when the hub cools, it will rest on the key and produce high stresses in the hub that could cause it to fail.

5.6.7 Couplings during mounting should be heated using a uniform heat source like resistance heating coils, Induction Heater, Oven etc. Open Flame Heating is not recommended, Distortion may occur due to localized spot heating or coupling may get stuck up due to non-uniform flame heating.

5.6.8 Do not exceed 600°F (315°C) during the heating of the coupling hub. Excessive Heat may soften the coupling due to annealing, Reducing the strength of the steel and may affect the performance characteristics of the Hub.

5.6.9 Special handling devices are required such as tongs, threaded rods placed in puller holes in the coupling hub, etc.

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5.7 Servicing & Inspection of inlet guide vanes

5.7.1 Remove old grease and dirt from the linkage assembly & their bearings/ bushes and inspect, then replenish the grease.

5.7.2 Conduct a visual inspection for corrosion, erosion and cracking of blades, damper seals & frame assembly.

5.7.3 For double inlet configuration, verify that the both side vanes are operating simultaneously. This is to be checked by establishing a reference dimension, actuating the vanes, and comparing the two sides for balance.

5.7.4 Check the linkage pins for wear & replace if required.

5.7.5 Check pin welding with flaps for any crack, soundness of weld and ensure smooth area change at pin and flap weld.

5.7.6 Check the IGV link assembly for any bolt looseness/ backlash and DPT of weld joints/ threaded parts in linkage assembly. Inspect the IGV actuator gear foundation for corrosion/ abnormality. Ensure IGV system /linkage is not fouling with any structure, which may restrict its free movements.

5.7.7 Components under Fatigue loading: - Fan components, which are under fatigue loading like IGV link assembly parts etc are to be replaced in 10 years, if having history of any failure.

5.7.8 There must be mechanical indicator for showing opening & closing position of the inlet guide vanes. Check & set the mechanical and electrical limits for operation.

5.8 Inspection of suction chamber and spiral casing

5.8.1 Inspect & repair the suction chamber & spiral casing for erosion.

5.8.2 Inspect the inlet cones fitted in suction chambers for erosion & measure the clearance (gap & overlap) between inlet cones & impeller. Maintain this clearance as it will deteriorate fan performance.

5.8.3 Inspect/ ensure the condition of shaft seal with casing and its clearance. Seals (floating type seals) for the shafts are provided to reduce the leakage through the gaps.

5.8.4 In some fans, wear liners are fitted inside the casing housing for longer life. Inspect these liners & replace based on condition. Also check the thickness of liners, if provided.

5.8.5 Inspect expansion joints at suction & discharge duct for cracks and holes after cleaning the flyash build up in the folds of the expansion joints. Suction side expansion joints severely affect the performance of the fan.

5.8.6 Inspection and replace the man door gaskets.

5.8.7 Cleaning & inspection of silencer suction manifold by compressed air. Also check the supports of silencer.

5.8.8 Check footings and supports for any cracks by DPT and corrosion sign.

5.9 Fan Internal Inspection during short shutdowns

5.9.1 Inspection to be done after cleaning the impeller assembly for erosions, if required thickness of impeller at various points may be measured from D meter.

5.9.2 Inspection for oil leakages from servomotor, hoses, bearing housing etc.

5.9.3 Attend any reported problem

5.10 Checks / Inspections during Long shut down /RSD

5.10.1 Once in 15 days, run the each lube oil pump for half an hour and operate the inlet guide vanes from full open to full close of blades.

5.10.2 Periodically exercise Fan's by rotating it several times and leaving it a different (90-degree) angle, if shut down is more than 15 days.

5.10.3 Testing of lube oil quality (Moisture & MI) by chemistry. Oil sampling for WDA analysis on Quarterly basis to be continued. Oil sample to be taken with lube oil pump in service.

5.10.4 The outside surfaces of the fan parts should be checked for corrosion. If found, then take anti-corrosion measures.

5.10.5 If vibrations of any fan are on higher side, then root cause analysis & action plan to be prepared after vibration analysis, while shutting down the unit.

6.0 HIGH FAN VIBRATIONS

The fans are prone to erosion, if fan is handling erosive media. This may cause unbalance on impeller. Vibration is the best indicator of overall mechanical condition and the earliest indicator of the developing defects. First of all, the vibration signature of a machine should be taken & analyze for possible causes of high vibration. The shape, size & location of unbalance weight may be decided for individual fan, however it should have rounded corners.

Most possible reasons of vibrations occur in fan units are listed below.

Guidelines for Inspection and Maintenance for Radial Fans

- 6.1 Bearing failures
- 6.2 Loose components
- 6.3 Imbalance
- 6.4 Worn/damaged parts
- 6.5 Misalignment or bent shaft
- 6.6 Improper Lubrication
- 6.7 Improper clearance of moving parts.
- 6.8 Excitation of a resonant frequency
- 6.9 Corrosion, erosion, high/low cycle fatigue effects
- 6.10 Flow induced vibrations from rotating stall
- 6.11 Electrical faults in motors
- 6.12 Loose foundation bolts
- 6.13 Other Foundation problems
- 6.14 Material deposition on the rotor /blade

7.0 Interlocks & Protections

- 7.1 Protections & interlocks of fans must be tested after maintenance work.
- 7.2 Approved Engineering limits for alarm / trip of fans on high vibrations & high bearing temperature should not be altered.
- 7.3 The following additional protections to be ensured for ID Fans (BHEL) in 500 MW units as per committee report ref- OS/Blr/CC/Fan dtd 28/04/2017.

To ensure flow thru each bearing Flow transmitters is the best option however in Rihand Stage I and Farakka stage II Pr transmitter have been provided by OEM. As pressure transmitters are readily available at station therefore same may be provided at first opportunity till flow transmitters are procured and installed in individual journal bearing inlet lube oil line after regulating valve.

Flow set values for 500MW unit ID Fans (NDZV 47S) - start permissive >10LPM, Alarm < 8LPM

Tripping of main motor when lube oil header pressure is less than 0.4 ksc & flow (DE or NDE bearings) is less than 2 LPM for more than 30 sec.

8.0 Special T&P for maintenance of Radial fans

Special T&P for maintenance of radial fan as given below, should be procured. Usage of the same must be ensured, wherever it is available.

Guidelines for Inspection and Maintenance for Radial Fans

SI No	Description	Quantity
1	Hydraulic Jack with pump, hoses and connectors of 50/75/100MT capacity	1 Set
2	Hydraulic Jack with pump, hoses and connectors of 200MT capacity	1 Set
3	Laser Alignment Kit with pre-cut SS Shims of different sizes	1 Set
4	Infra-red Non-contact Thermometer	1 No
5	Portable Vibration Analyzer	1 No
6	Special Tool Kit for Fans (As per OEM List)	1 No for each model Fan
7	Bearing Induction Heater	1 Set
8	Hydraulic Pullers	1 Set of each capacity
9	Pneumatic/ mechanical Torque Wrenches	1 Set
10	Bore gauge, Micrometre, Vernier calliper	1 Set
11	Hydraulic nut / Hollow jack as per requirement	1 Set
12	Master level	2 No
13	Trolley mounted Portable Compressor	1 No
14	Portable Oil Filtration Unit	1 No
15	Arrangement (Jig/fixture) for rotation of fan during reserve shut down/long shutdown	01no for each type of fan
16	Resistance Heating Coil	1 No
17	Heating Torch (For large couplings)	2 Nos

9.0 This LMI to be reviewed in frequency of 02 years

Guidelines for Inspection and Maintenance for Radial Fans

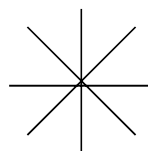
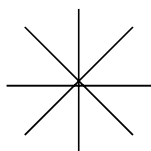
8. Bearing Clearance (Sleeve Bearing)

Bearing size	Axial Gap		Side Clearance		Oil Clearance	
	Min	Max	Min	Max	Min	Max
As Per Drawing						
DE						
NDE						

9. Impeller Seal Clearance (Impeller to inlet cone clearance)

DE

NDE



As per Drawing

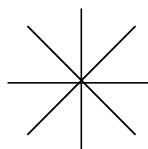
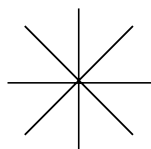
Top :

Bottom :

10. Impeller Seal Over Lap (Impeller to inlet cone penetration)

DE

NDE

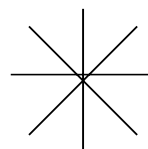


As per Drawing

DE :

NDE :

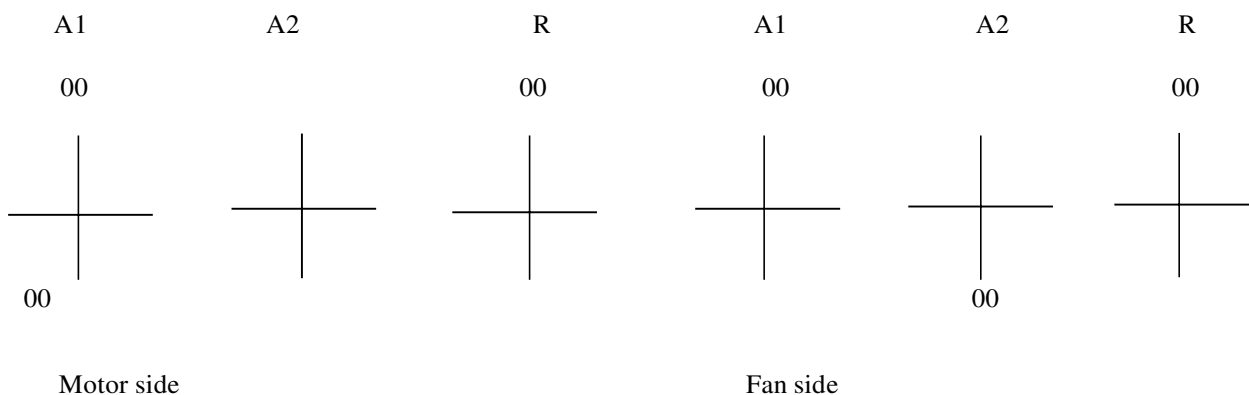
11. Back Plate Clearance



As per Drawing:

Guidelines for Inspection and Maintenance for Radial Fans

12. Coupling Alignment



Coupling Gap

Description	Drive end
As per drawing	
Actual	

13. Servicing of IGV / Damper Bearing :
14. Smooth operation of IGV Damper :
15. Calibration of IGV / Damper w.r.t Actuator :
16. Cleaning of filters (Hyd. coupling & LOS) :
17. Cleaning of cooler (Hyd. coupling & LOS) :
18. Cleaning of tank :
19. Hydro test of cooler (Hyd. coupling & LOS) :
20. General cleaning :
21. Lub oil pressure (Hyd. coupling & LOS) :
22. Oil Replacement
23. Calibration of all instruments (Thermometer, Pressure gauge, Pressure switches etc :
24. Scoop calibration :
25. Greasing of connecting coupling :

During Trial Run

1. Vibration Readings

Date : Time :

Scoop position : Motor current :

Location	FNDE	FDE	HCDE	HCNDE	MDE	MNDE
Horizontal (D/V)						
Vertical (D/V)						
Axial (D/V)						

2. Bearing Temperature

FDE	FNDE	MDE	MNDE	Hyd. coupling working oil	Working oil pressure

Check list of Fan trial after major maintenance

Date:-

- | | | |
|-----|---|------------|
| 1. | All protections and interlocks checked | Yes / No |
| 2. | All tools, personnel and debris removed | Yes / No |
| 3. | All access doors closed | Yes / No |
| 4. | Lub Oil System Parameters | |
| | Lub oil level in the tank. | Ok/ Not Ok |
| | Any oil leakage | Yes / No |
| | Filter DP | |
| | Lub oil pump discharge pressure | |
| | Fan DE/NDE bearing oil pressures | |
| | Motor bearings oil pressure/ level | |
| | Voith coupling oil pressure (if given) | |
| | Oil flow in view glasses in bearing return oil lines. | Ok/ Not Ok |
| | Cooling water lines valves opened (cooler/bearing) | Yes / No |
| | Water flowing through the view glass of lines | Ok/ Not Ok |
| 5. | Position of Inlet & outlet dampers | Ok/ Not Ok |
| 6. | Coupling guards fitted | Ok/ Not Ok |
| 7. | Direction of rotation of the motor. | Ok/ Not Ok |
| 8. | Vibration of fan & motor at local | |
| 9. | Temperature of the fan/ motor bearings | |
| 10. | Any abnormal noise | Yes / No |

Signature: -

Name: -

Designation: -

Deptt- Boiler Maintenance

Signature: -

Name: -

Designation: -

Deptt- Operation

Annexure-III

2 Form for Rotary Machines

BOILER ROTARY PARTS FAILURE REPORT (C-2 FORM) (TO BE FILL WITHIN 28 DAY OF INCIDENTS)										C1 Submission date		
										Financial year/ Station/Unit/ Area/Date		
1 Station	2 Unit No.	3 Failure Date								4 Last Unit Overhaul		
5 Last Unit Start Up		6 Total RP Failures in FY		7 Failures in Same Equip Since Last OH								
8 Total No of Similar failures in Unit in last 10 Yrs		9 Equipment S/d in Hrs, Generation Loss in Mus & Financial loss in lacs										
10 Equipment Health After Last Overhaul (Temp, Vib, Noise)												
11 Equipment Health Just before Failure ,S/D (Temp, Vib, Noise and last WDA)												
12 Equipment Health after Restoration (Temp, Vb, Noise)												
EQUIPMENT FAILURE DETAILS												
13 FAILED PART MARKED ON UNIT ASSEMBLY DIAGRAM (insert here)												
DESCRIPTION OF FAILED EQUIPMENT												
14 PHOTOGRAPHS OF LOCATION OF FAILURE BEFORE START OF REPAIR WORK (insert here)												
15 DETAILS OF ALARM & PROTECTION IIS AND VALUES AT WHICH OPERATED (ADD CONTROL CURVES)												
16 DETAILS OF REPAIR WORK												
17 DETAILS OF INSPECTION OF OTHER SIMILAR EQUIPMENTS												
18 HISTORY OF Work done IN FAILED EQUIPMENT LOCATION IN PAST FIVE YEARS												
19 FAILURE MECHANISM												
20 INITIAL OBSERVATION AND ROOT CAUSE												
21 ACTION PLAN FOR ROOT CAUSE MITIGATION AND COMPLETION DATE												
22 ACTION PLAN FOR ROOT CAUSE MITIGATION IN OTHER UNITS W												
23 ODI/OGN/OIN/COS recommendation with respect to above failures												
24 STATION COMPLIANCE SCHEDULE WRT TO RECOMMENDATIONS: SOW BOQ NUMBERS AND TIME LINE												
FORMAT NO. OS/BLR/RP/002 REV.03										(Digital Signature Of Committee Head)		DATE
										NAME:		
										DESIGNATION:		
										Mobile No		
C2 copy in email to be provided to all concerned at station, region, COS KT member and Head COS Boiler												
ALL FAILED PART SAMPLES TO BE SEND TO COS FOR FURTHER ANALYSIS IF REQUIRED												
C2 file to be closed by COS after root cause mitigation action plan BOQ finalisation by station and its inclusion in SOW.												

Annexure-IV

Health Matrix for Fans

Rotary Parts Equipments Health Assessment and Replacement Monitoring Matrix																
Station																
Unit																
Equipment	Date of Last Bearing Replacement	Date of last inspection/defects	WDA Report of Lub Oil		Bearing Vibration Report							OEM Recommendations		Bearing temperature		Availability of Spare assy(Y/N)
					Online (Control Room Reading)				Offline measured by Condition monitoring (Location.....)							
			Value	Date	DE	Date	NDE	Date	Radial H & V	Axial	Date	Alarm	Trip	DE	NDE	
PA Fan A																
PA Fan B																
ID Fan A																
ID Fan B																

Guidelines for Inspection and Maintenance for Radial Fans

Check list For LMI Guidelines for Inspection and Maintenance for Radial Fans

S No	Details	Responsibility
1	During each overhaul both suction cone of radial ID fan to be pulled back from its original position to check the condition of weldment of impeller mounting flange and cone sitting rings.	HOD (BMD)
2	Visual inspection of all shaft and impeller welds during major overhauls including the balance weight welds. They should be inspected for any signs of obvious distress and to assess the extent of any wear.	HOD (BMD)
3	All the shaft welds should be inspected for defects like cracks, undercuts, notches, etc. by either by DPT or magnetic particle test. -	HOD (BMD)/HOD(FQA)
4	Dye penetrant test or Ultrasonic testing of sleeve bearings (lower & upper half) must be carried out for checking cracks & bonding of Babbitt metal.	HOD (BMD)/HOD(FQA)
5	Clearance & blue match of Journal/ sleeve bearings with shaft journals must be checked as per procedure given by OEM	HOD (BMD)/HOD(FQA))
6	Digital drawing to be developed for emergency requirement spares, which are not available in stock.	HOD (BMD)/HOD(MTP)
7	Inspection of IGV actuator gear foundation for corrosion/ abnormality.	HOD (BMD)/HOD(C&I)
8	Quarterly oil monitoring through wear debris analysis (WDA) must be followed strictly. The safe limit of PQ index for fans is 25 (As given in NETRA WDA report). The moisture content should be less than 100 PPM or as recommended by OEM. Oil should be clear and transparent and should not be hazy. Centrifuging may be done to remove moisture. In case of any abnormality in the WDA, necessary corrective action as follows should be taken. Oil sample should be collected from system before filters.	HOD (BMD)/HOD(MTP)/ HOD(CHEM)
9	Protections & interlocks of fans must be tested after maintenance work.	HOD(Oprn)